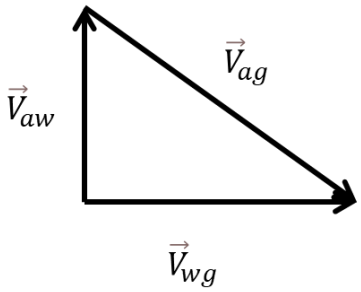
Quiz#5

Section: _____ **Group:** _____

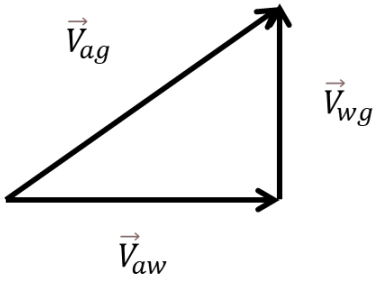
Name(print): _____ **Student ID:** _____

1. Given an airplane's velocity relative to the wind as $\vec{V}_{aw}$, the wind's velocity relative to the ground as $\vec{V}_{wg}$, and the airplane's velocity relative to the ground as $\vec{V}_{ag}$, identify the correct vector tip-to-tail triangle representing the relationships between $\vec{V}_{aw}$, $\vec{V}_{wg}$ and $\vec{V}_{ag}$.

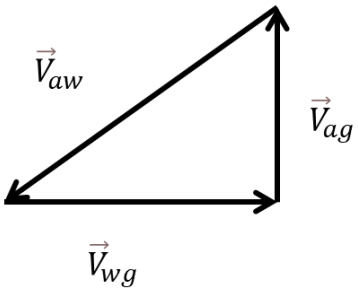
A



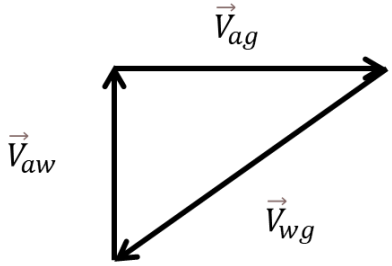
B



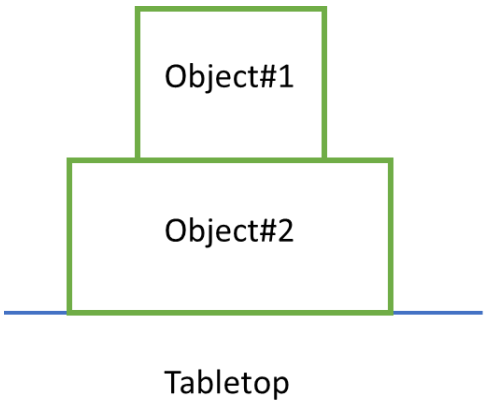
C



D



2. Two stacked boxes as shown below. According to Newton's third law, which of the following statements is a pair of Newton's third law forces?



A. The normal force object#2 exerting on object#1 and the gravity of object#1
B. The gravity of object#1 and the downward push object#1 exerting on object#2.

C. The normal force object#2 exerting on object#1 and the gravity of object#2.
D. The normal force object#2 exerting on object#1 and the downward push object#1 exerting on object#2.

Using the five-step problem-solving strategy, solve the following problem:

3. A block with a mass of 2.1 kg is pushed across a horizontal tabletop by a force of 15 N as shown below. The coefficient of kinetic friction between the block and the tabletop is 0.28. Determine the magnitude of the acceleration of the block.

